ITM SKILLS UNIVERSITY

Department of Computer Science & Engineering

SOFTWARE ENGINEERING & PROJECT MANAGEMENT

PETPAL

Case Study No. 64

RISK REGISTER & CLOSURE NOTE

7-Point Risk Register, 5×5 Probability-Impact Matrix, RMMM Plans & Formal Project Closure

SUBMITTED BY

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Project: PetPal — Appointment Booking and Records App for a Veterinary Clinic Chain
Case Study: No. 64
Subject: Software Engineering & Project Management

SECTION 1 & 2

1. Comprehensive Risk Register (7 Key Risks)

TABLE 1.1: PROJECT RISK REGISTER

ID	RISK DESCRIPTION	PROB (1-5)	IMP (1-5)	SCORE	RMMM MITIGATION & CONTINGENCY STRATEGY
R1	Existing EMR Integration Sync Failure	4	5	20 (HIGH)	Mock sandbox early testing; asynchronous queue with CSV batch fallback.
R5	Scope Creep (Pet-Food Store Request)	4	4	16 (HIGH)	Formal 6-step CCB change control; deferral to Phase 2 roadmap.
R2	Vaccination Reminder Daemon Failure	3	5	15 (HIGH)	Redundant cron schedulers, dead-letter queue, and manual replay dashboard button.
R3	Payment Gateway Webhook Timeout	3	4	12 (MED)	Polling fallback reconciliation worker and HMAC signature validation.
R6	Appointment Slot Concurrency Conflict	3	4	12 (MED)	Atomic 10-minute temporary slot reservation hold with DB unique constraints.
R4	Data Privacy / RBAC Boundary Breach	2	5	10 (MED)	Strict token claim validation, salted bcrypt hashing, and TLS 1.3 encryption.
R7	Notification Delivery Carrier Drop	3	3	9 (MED)	Multi-vendor fallback SMS/Email routes.

SECTION 3 & 4

2. Cohesion & Coupling & 3. Project Closure

High Functional Cohesion: Each subsystem (Pet Management, Appointment Booking, Vaccination Reminder, Payment, Notification, Dashboard) encapsulates exactly one business domain.

Low Coupling: Subsystems interact via stateless REST APIs and asynchronous message queues without direct database sharing.

Project Closure Statement: All technical specifications, architectural designs, CPM schedules, SQA test plans, and high-fidelity UI prototypes have been 100% completed and verified in accordance with academic standards.